

Abstract

Dexterous manipulation relies on accurate motion-direction information for coordinating finger movements and correcting errors. We present a haptic device that uses a moving electromagnet and a finger-worn magnetic puck array to generate traveling normal indentations across the skin, creating directional tactile motion. We evaluated the device in a two-alternative forced-choice study in which participants identified leftward or rightward motion across different actuation conditions. Initial results suggest that users can reliably distinguish the rendered motion direction.

Motivation

- Fingertip haptic interfaces can enrich interaction by delivering spatial tactile information directly to the skin. Magnetic actuation provides a controllable way to generate localized skin deformation without placing an active actuator directly on the skin.

Research Question

- Can magnetic fields provide tactile haptic feedback for surface texture rendering?

Haptic Rendering

- As the electromagnet moves along the arc, the peak normal displacement shifts across the puck array, producing a traveling indentation on the fingertip.

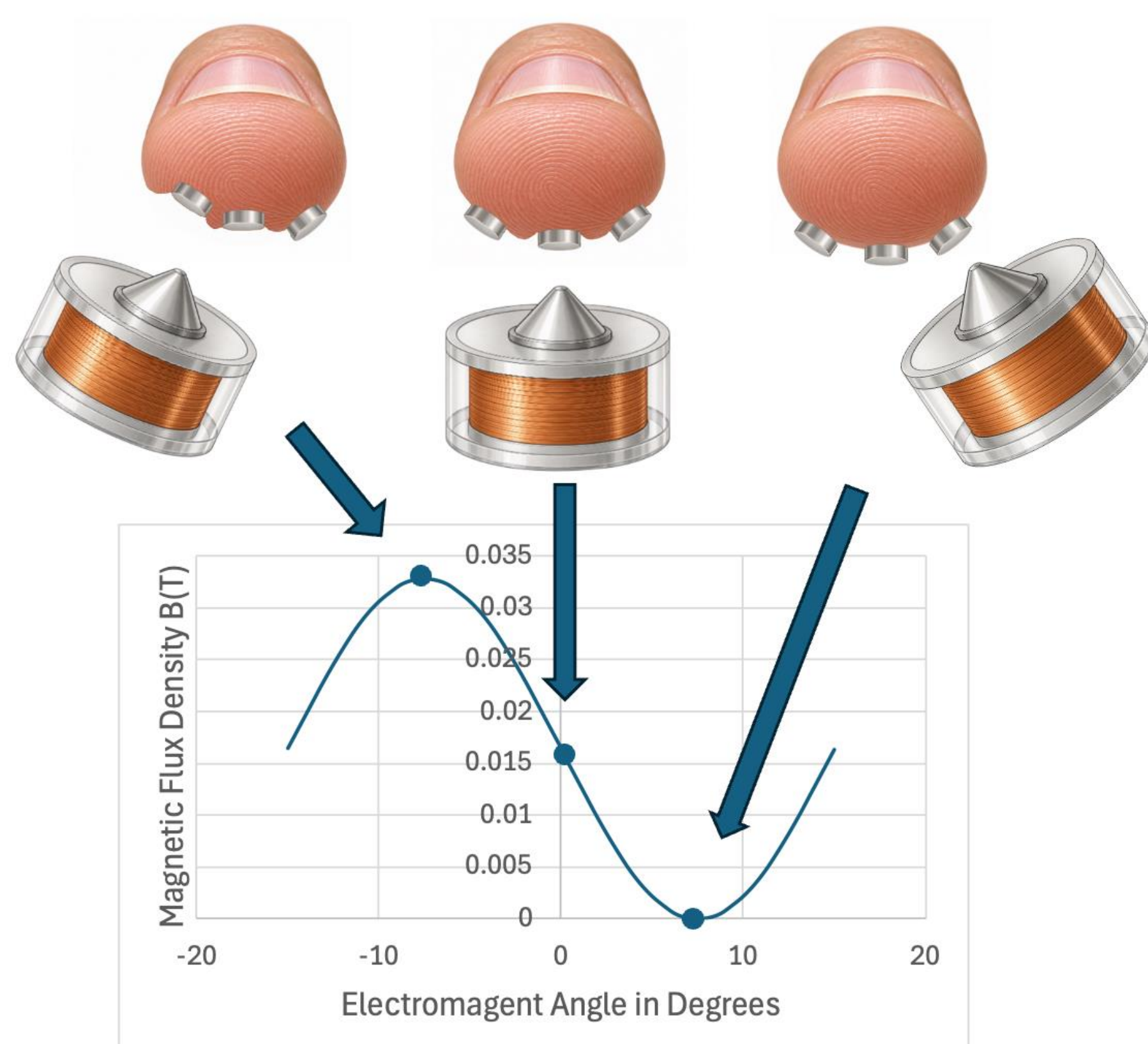


Fig. 1 Haptic rendering mechanism: spatial modulation of electromagnetic field with traveling electromagnet
Magnetic flux density is measured 3mm above the tip of the core

Design

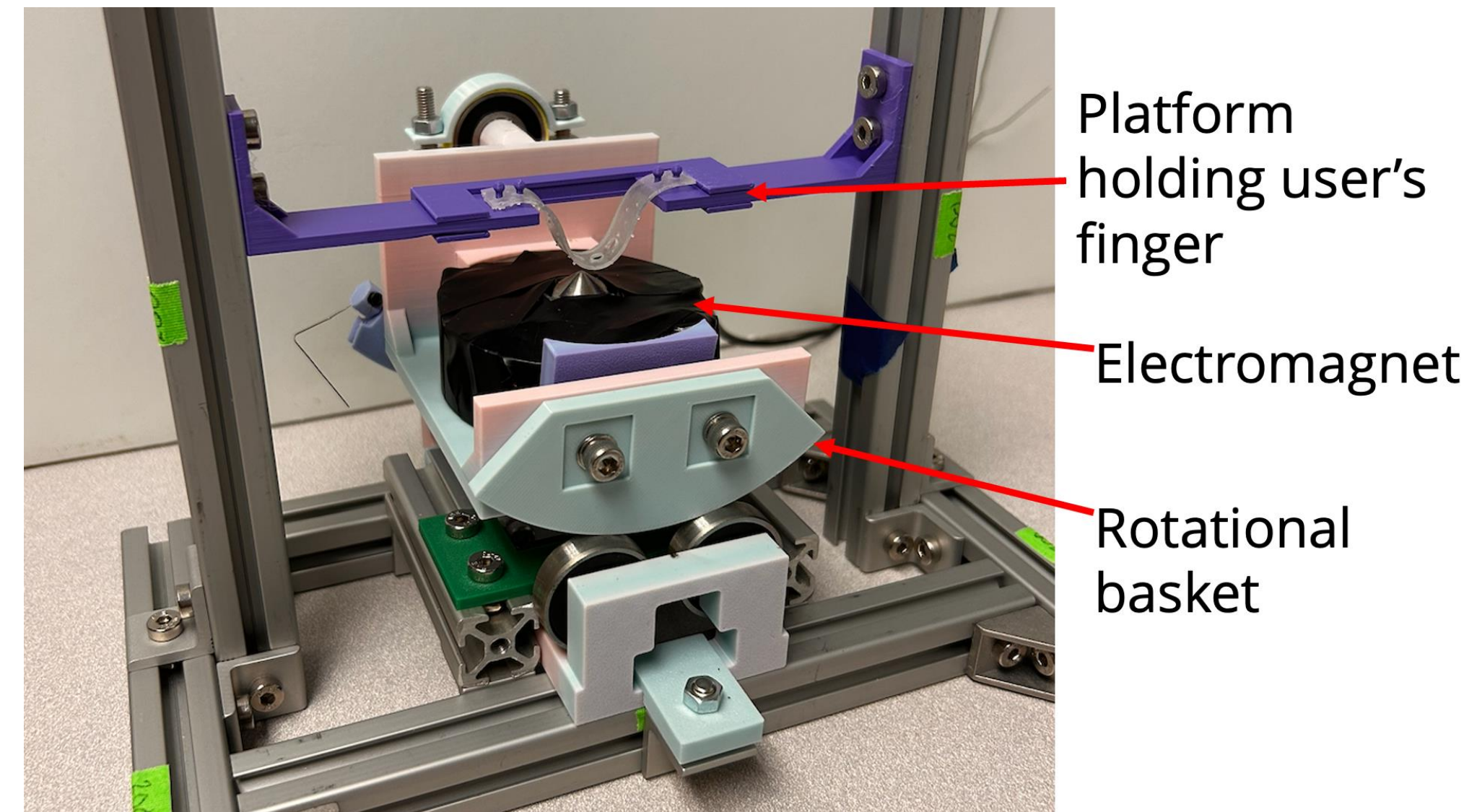


Fig. 2 Device component description

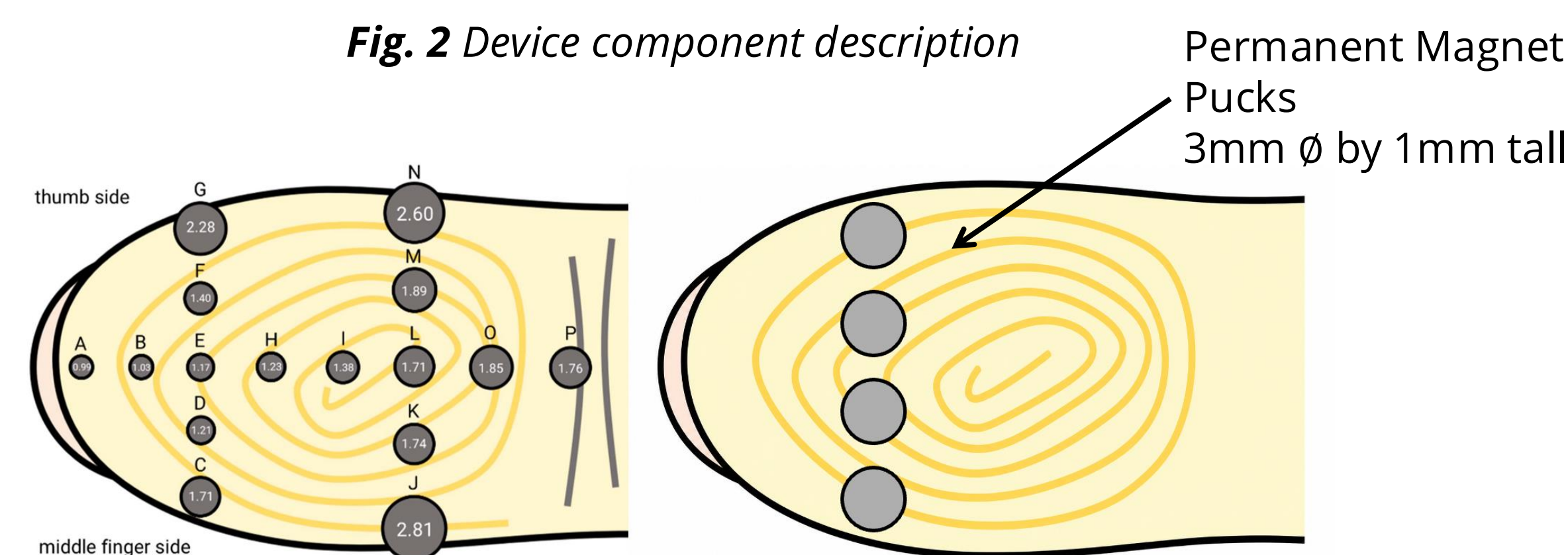


Fig. 3 Magnet pucks placing was based on findings from Sobue et al. (2025)
Magnets are placed on the finger using a resin strap and elastic tape.

Experiments

User Study

- We conducted a 2AFC study with 12 participants to test left-right tactile direction perception.
- Each participant completed four conditions: 2 speeds $\times$ 2 actuation profiles, with 30 trials per condition.
- A total of 1,440 trials were collected, recording direction response and reaction time.



Fig. 4 Illustration for User Study

Acknowledgements and Reference

This work was supported by the National Science Foundation (NSF) under NSF HCC Grant # 2312153 and the NSF Human Augmentation via Dexterity Engineering Research Center (HAND ERC).

References: [1] M. Sobue et al., "Measuring the Distribution of Tactile Acuity at the Index Finger and Thumb Fingertips," *IEEE Trans. Haptics*, 18(3), 663–670, 2025.

Results

Participants showed measurable direction discrimination, with an overall $d' \approx 1$.

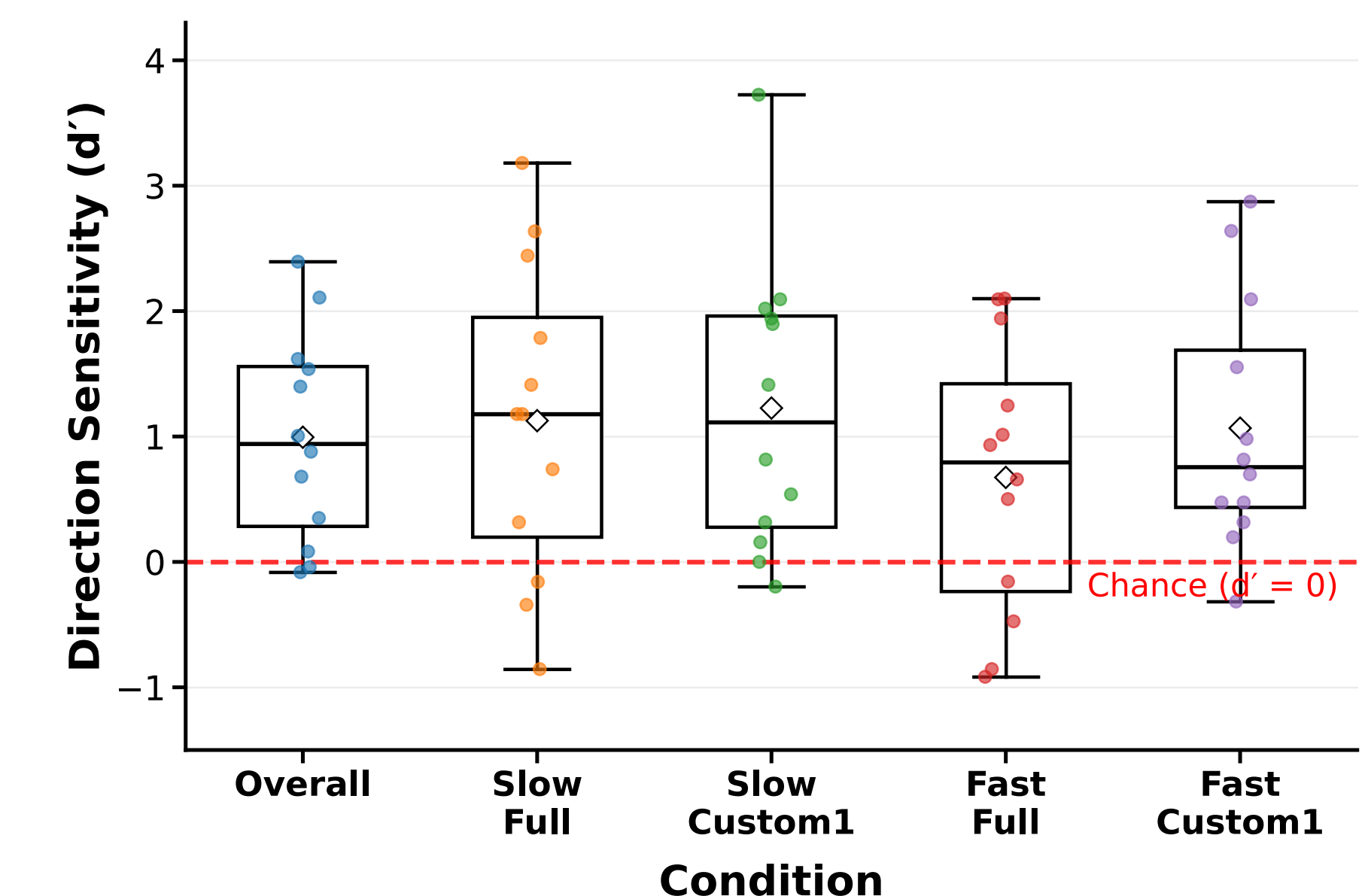


Fig. 5 Distribution of Direction Sensitivity Across Conditions

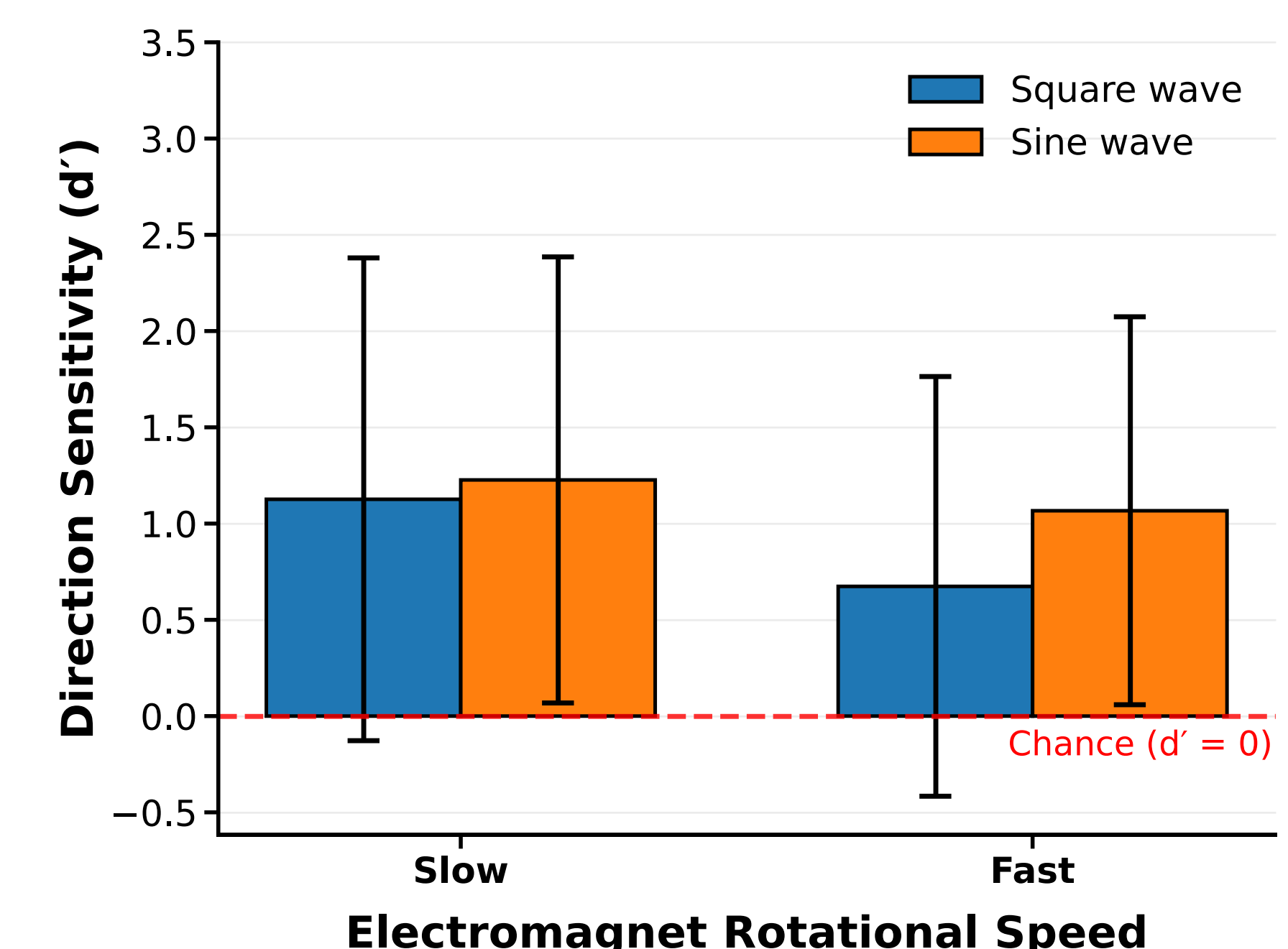


Fig. 6 Distribution of Direction Sensitivity Comparison

Contribution

- Design of a magnetic haptic device with a moving electromagnet and magnet puck array.
- Directional tactile rendering through electromagnetic actuated traveling normal indentation.

Conclusions and Future Work

- Device can produce tactile feedback in the form of traveling waveforms.
- Participants showed measurable left-right direction discrimination, with an average $d' \approx 1$.
- Integrate force sensing for normal force relation to direction-identification performance.
- Conduct threshold experiments for minimum stimulus discrimination.